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New Product Development Practices and Early-Stage Firms: Two In-Depth Case Studies

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New product development practices (NPD) have been well studied for decades in large, established companies. Implementation of best practices such as predevelopment market planning and cross-functional teams have been positively correlated with product and project success over a variety of measures. However, for small new ventures, field research into ground-level adoption of NPD practices is lacking. Because of the risks associated with missteps in new product development and the potential for firm failure, understanding NPD within the new venture context is critical.

Through in-depth case research, this paper investigates two successful physical product-based early-stage firms' development processes versus large established firm norms. The research focuses on the start-up adoption of commonly prescribed management processes to improve NPD, such as cross-functional teams, use of market planning during innovation development, and the use of structured processes to guide the development team. This research has several theoretical implications.

The first finding is that in comparing the innovation processes of these firms to large, established firms, the study found several key differences from the large firm paradigm. These differences in development approach from what is prescribed for large, established firms are driven by necessity from a scarcity of resources. These new firms simply did not have the resources (financial or human) to create multi- or cross-functional teams or organizations in the traditional sense for their first product. Use of virtual resources was pervasive. Founders also played multiple roles concurrently in the organization, as opposed to relying on functional departments so common in large firms. The NPD process used by both firms was informal—much more skeletal than commonly recommended structured processes. The data indicated that these firms put less focus on managing the process and more emphasis on managing their goals (the main driver being getting the first product to market). In addition to little or no written procedures being used, development meetings did not run to specific paper-based deliverables or defined steps. In terms of market and user insight, these activities were primarily performed inside the core team—using methods that again were distinctive in their approach. What drove a project to completion was relying on team experience or a "learn as you go approach." Again, the driver for this type of truncated market research approach was a lack of resources and need to increase the project's speed-to-market.

Both firms in our study were highly successful, from not only an NPD efficiency standpoint but also effectiveness. The second broad finding we draw from this work is that there are lessons to be learned from start-ups for large, established firms seeking ever-increasing efficiency. We have found that small empowered teams leading projects substantial in scope can be extremely effective when roles are expanded, decision power is ground-level, and there is little emphasis on defined processes. This exploratory research highlights the unique aspects of NPD within small early-stage firms, and highlights areas of further research and management implications for both small new ventures and large established firms seeking to increase NPD efficiency and effectiveness.

Introduction

New product development (NPD) processes, best practices, and management methodologies have been the focus of intense research over the last several decades. Seminal research argues the importance of practices ranging from the implementation of cross-functional teams (Taguchi and Nonaka, 1986; Wheelwright and Clark, 1992) to the planning and adoption of scalable architectures and product platforms (Meyer and Lehnerd, 1997) on improvements in effectiveness and efficiency in project outcomes. The NPD process is arguably the most important dynamic capability within a firm (Nelson, 1991). Overwhelmingly, this research has been targeted toward established corporations with existing product portfolios.

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But what about new product development within early-stage, small entrepreneurial firms? Of the 25 million businesses in the Unites States, approximately 99.7% are considered small (U.S. Small Business Administration, 2004). Typically, small businesses are comprised of fewer than 100 people. Research has shown that these firms are key contributors to innovation and economic growth (Carree and Thurik, 2003), as small businesses produce 51% of the of the U.S.'s private GDP and 47% of business sales (U.S. Small Business Administration, 2004). Small firms are hotbeds of innovation, creating over 13 times more innovation per research employee than large companies (U.S. Small Business Administration, 2004). Eight hundred fifty thousand small, new businesses are created each year (Price, 2006), and studies suggest that roughly one in nine people in the United States is working to start a new business (Acs, Arenius, Hay, and Minniti, 2004). Data suggest that effective management of the development of new products in such firms is no less important—and perhaps more so—than for large, mature corporations, given historically high product and firm-level failure rates (Adams, 2004; Headd, 2003; Song, Podoynitsyna, van der Bij, and Halman, 2008). In new firms, the successful launch and commercialization of the first product has been shown to be critical to survival and is perhaps the most important aspect of a firm (Schoonhoven, Eisenhardt, and Lyman, 1990). Approaches to and successful outcomes from new product development are one of the foundations for venture survival (Gartner, Starr, and Bhat, 1998). Understanding the product development processes and methods used to increase the likelihood of successful outcomes is paramount to the new venture. Small, new firms do not have the resources—time, money, and staff—to absorb product failure that is the result of a poorly executed development process. Unfortunately, it has been argued that entrepreneurs can be long on innovativeness and technical skill but short on management process knowledge (Cooper, 1970; Oakly, 2003; Saxenian, 1985) to the detriment of long-term success. As such, understanding NPD practices in the context of new ventures is fundamental to entrepreneurial management knowledge.

In large established firms, certain NPD practices such as up-front market planning and adoption of cross-functional teams have been associated with successful outcomes (Adler, 1995; Cooper, 2001). But how does the ground-level adoption of NPD practices in new ventures differ from that of established firms? Are heavily researched practices such as predevelopment market planning and use of formal processes such as phase or Stage-Gates® even appropriate for the small nascent firm? This research seeks to investigate links between NPD literature for established firms and field implementation of best practices at new ventures developing physical products as it can be argued that the bulk of NPD research has focused on assembled goods (from automobiles to computers). Are small, early-stage firms using commonly prescribed established firm methods such as phased development, predevelopment market planning, and true cross-functional teams, and can insights be drawn that correlate these NPD factors to outcomes? Are new ventures different in their approach to NPD? Are there any insights from new ventures that might be applied to innovation creation in large, established firms?

In order to accomplish this task, an in-depth investigation was performed over a multiyear period on two small early-stage firms developing multipart, physical consumer products. Case study methodology was chosen as a way to intimately understand the environment and the relevant factors affecting NPD within these firms. This approach has been used effectively to describe a wide variety of subjects within NPD, including development cycle time (Mabert, Muth, and Schmenner, 1992), cohesiveness among product teams (Brockman, Rawlston, Jones, and Halstead, 2010), and specific design

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issues such as brand development and recognition (Karjalainen and Snelders, 2010). Jackson (2000, p. 238) notes: "based on first-hand fieldwork, qualitative methods convey and understand the inner life and texture of a particular social group or locality." In this research, each firm was followed from development through commercialization and ongoing sales.

This investigation is valuable for several reasons. First, the research adds to the burgeoning field of entrepreneurship research in managerial methods and practices that can increase firm success. Recent works such as Heirman and Carysse (2007) and Song et al. (2008) have noted the importance of collaborations with third parties, founders' marketing experience and approach, and supply chain integration. Our lens is focused on the development process itself, which Gartner et al. (1998) noted as one of the foundations for venture survival. Next, large, established firm innovation paradigms are compared to small new ventures—highlighting potential differences and similarities that can foster improvement in best practices for both spaces. In the next section, the rationale behind the selection and applicability of NPD practices to the new venture are discussed. Study results are then presented with a discussion of the main findings. The paper concludes with the identification of key managerial implications and directions for future research.

NPD Practices and Research Methodology

Developing new products is costly and risky. Multipart consumer product development costs can range from $50,000 to millions of dollars depending on complexity and technology newness (Meyer and Marion, 2010). Extremely complex products such as automobiles can exceed well over $1 billion in R&D costs (Mol, 2001). Unfortunately, despite intense research and development of new NPD methodologies, the failure rate for new products and services remains elevated at approximately 40% (Adams, 2004). For small early-stage firms, the stakes are even higher, given the cost of product development and risk for product and firm failure (50% of new ventures cease operation after four years [Headd, 2003]). Since literature on best practices for NPD in early-stage firms is sparse at best, key findings of the long, well-established stream of NPD research for large, mature corporations are used as a proxy for this in-depth investigation. Robert Cooper's seminal work (2001) outlines two fundamental aspects for mitigating the associated risks of commercializing successful products. These fundamentals are: (1) doing the *project right*, based on common success factors among successful NPD companies including cross-functional teams, up-front market planning, and early product concept definition; and (2) doing the *right projects*. Cooper argues that product selection and product planning methodology are essential to a successful product launch and life cycle.

The important question to be asked is what do Cooper's fundamentals and associated "methods and best practices" mean for the small new venture? What is their relevance to efficient and effective outcomes in this space? In order to begin to investigate those questions, the reality of new venture creation and relation of these situational drivers to associated large, established firm NPD practices needs to be explored. Balachandra and Friar (1997) found that NPD is contextual, arguing that among different industries, technologies, and levels of innovation that the management process of NPD differs widely between sectors. It can be argued that NPD is also contextual in small early-stage firms, with unique differences between technologies and industries (e.g., consumer devices versus biotechnology, etc.). A software firm rapidly prototyping a new social networking site can experience very different development milestones than a biotechnology firm developing a new gene therapy (Friar, Marion, and Kinnunen, 2009). That is why our focus here is on a particular industry and one context of NPD: technology-based new ventures that develop multipart, physical products that require assembly. Specifically, our research is centered on hand tools and office machines. This industry requires that products be conceptualized to meet an obvious or latent market need, go through a design process and testing, source manufacturing, and establish a supply chain to fulfill orders. We argue that this industry most clearly represents traditional, established firm NPD as described in the works of Cooper (2001, 2005) and Ulrich and Eppinger (2004).

In our industry focus, small start-ups in fact experience similar project needs throughout the development process as large established firms, pointing to the importance of NPD practices and processes in these new firms (Gartner, 1985; Gartner et al., 1998; Schoonhoven et al., 1990). A project can be defined as a unique set of tasks with a beginning, an end, and a well-defined outcome (Project Management Institute [PMI] Standards Committee, 1996). A start-up is in essence a project, which starts with an entrepreneur's idea, obtains funding, follows on agreed milestones during design and development, and ends through ongoing operations, closure, or a liquid event (Loch, Solt, and Bailey, 2008). Ulrich and Eppinger (2004, p. 26) state: "new ventures or start-ups are among the most extreme examples of project organizations: every individual, regardless of function, is linked together

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by a single project—the growth of the new company and the creation of its product(s)." While a start-up is a project, and there are clear indications that traditional NPD practices are applicable to these firms, distinctions between large, established firms and the small new venture must be explained.

There is an obvious difference between a project at a well-funded firm such as Apple and a project undertaken at a new venture. Resources can be extremely limited at the new firm—in both funds and personnel—and this dictates how the new firm navigates through the process toward commercialization of the first revenue-generating product. Most new ventures start with one to three founders (Adams, Almeida, and Ferreira, 2009). This presents a challenge for the firm that needs to design and implement a new product that requires any level of engineering, manufacturing, and launch. Additionally, funds can be scarce. Zimmerer and Scarborough (2008) note the difficulty of raising capital for new ventures. They state (p. 463): "most entrepreneurs face difficulty in finding outside sources of financing . . . venture capitalists have become risk adverse and private investors remain cautious. The result has been a credit crunch for entrepreneurs looking for small amounts of start-up capital. Those needing between $100,000 and $3 million are especially hard hit because of the vacuum that exists at that level of funding." In the industry under investigation, those levels of funds exactly correlate with the costs of product development. Once investment has been garnered, these firms work under a finite, declining budget—not having the luxury of continuous cash flow generated by a product portfolio. As such, resource management of capital (both human and financial) is paramount from concept to successful commercialization and may have an impact on how NPD practices are adopted at these firms.

Following the project-related issues detailed above for new ventures developing physical products, our research is aligned to Cooper's (2001) study which highlights critical characteristics of NPD. These include factors such as the use of cross-functional teams, a formal development process for project control, and use of predevelopment market planning. In the next section, the applicability and influence these practices may have on the new venture are explored, and the expected differences between new and established firms are highlighted.

NPD Practices and Links to Early-Stage Firms

The next section discusses best practices and their appropriateness, or lack thereof, to the needs of new ventures. This literature review is focused on NPD within the context of the development and commercialization of physical, assembled products. Physical products arguably demand the greatest level of NPD practice adoption because of the need for capabilities in design, cost engineering, manufacturing, and supply chain management. In his 2001 study, Cooper lists 15 critical success factors for product innovation. These include ensuring product differentiation, having strong market orientation and attractiveness, doing predevelopment homework, having the proper organizational structure supported by strong management support, and implementing a phased development process. In Cooper's 2001 study, he synthesizes his 15 critical success factors into two succinct points, that firms must do the *right projects, correctly*. In this research, the similar approach is used to synthesize and categorize critical factors for improving NPD effectiveness—NPD effectiveness in terms of successful launch and life cycle being critical to the new venture. The process of doing the projects *correctly* is encompassed in two categories (having a cross-functional organization and predevelopment market planning/homework). Because Cooper's work in 2001 and 2005 in the main points to the use of a gated development process to help select and manage the *right* projects, our third category is the use of a formalized development process (e.g., a phase or Stage-Gate). Within these three categories, several drivers are investigated that describe the adoption within each main category. In the next section, we review appropriate literature that builds into these three categories, main drivers, and associated research questions.

The implementation of a cross-functional organization. Cooper (2001) noted that a cross-functional organization is a key ingredient to successful product innovation. Research has argued that increasingly established companies employ cross-functional teams to develop new products (Adler, 1995; Griffin and Hauser, 1996) and are finding positive effects on cycle time and project performance (Barczak and Wilemon, 2003; Taguchi and Nonaka, 1986; Wheelwright and Clark, 1992). Examples of cross-functional teams used in mature companies for the development of assembled and nonassembled products, systems, software, and services can be found in Smith and Reinertsen (1997) and Meyer (2007). Cooper (2005) defines cross-functional teams as those that have full-time team members from sales, engineering, marketing, and operations. Cooper (2001) also highlights the importance of resources and having them in place. For the small new venture, given their nature of

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small founding teams and limited financial resources to hire full-time personnel, the core team of founders is often bolstered by external resources, which have to perform a variety of functions from design to manufacturing (Marion and Meyer, 2011). This is particularly true in firms developing physical assembled products where team members can include a variety of engineers and designers, often globally sourced and virtually managed (Eppinger and Chitkara, 2006). New ventures with scarce resources will rely on a limited number of full-time members that have multiple responsibilities and will also rely on part-time external members to reduce capital expenditures. Written formally:

Proposition 1. New ventures will not implement a traditional cross-functional organization.

The formality of the NPD process. Cooper (2001, 2005) and McGrath (1995) have popularized the use of phase-gate systems to reduce cycle time and development expense in established corporations. The phase-gate comprises a set of sequential processes—invention, scoping the business opportunity, building the business case, developing product, testing and validation, and launch—and at the end of each management can decide whether to proceed to the next stage. These decision gates have appeal for some corporations because they provide points at which senior executives can veto forward progression of poor ideas and direct resources to better ones. For established firms, Cooper (2001) recommends a five-stage/phase process with decision gates at each stage. Research suggests that approximately 60% of established firms use some form of phased development approach (Cooper, 2005).

The overarching benefit described by supporters of phase-gate systems is that firms can select promising concepts up-front and stop less attractive ones from wasting resources during development. Management can essentially halt projects before reaching the market. In this study, three characteristics are investigated that describe the formality of process through the lens of Cooper (2001). Is there a formal procedure to stop an unattractive product from moving forward through the development process? Cooper (2001) calls this a go/kill process at some point in the design process. The second facet in our investigation of process formality is the use of official development stages and gates. Does the new venture use a formalized process—with a methodology that helps with project planning and control? For less complex or new ventures, Cooper (2001) proposed the truncated *fast-track* process for this purpose (now Stage-Gate Xpress®), which has three stages and two gates—and more recently a Stage-Gate Lite® with one gate and two main phases for simple projects. Given the context of limited human and financial resources, new ventures will have fewer decision gates, but at least one go/no-go prior to investing into producing the units and purchasing tooling, allowing the management team to exercise control over precious funds, without having to resort to onerous procedures. These firms will also have no formal phases, instead focusing on distinct project milestones. Finally, the adoption of project management tools such as task lists and Gantt charts is investigated. Has the project been managed using tools that foster planning and control? Cooper (2001) notes that a well-planned project will stand a better chance of being performed *correctly*. This project-based approach is in line with recent literature such as Loch et al. (2008), which examined management approaches in dealing with uncertainty in new ventures, illustrating how process can help entrepreneurial managers navigate unknowns. For those firms developing physical products, a more structured process of clearly defined and sequential phases can add clarity and stability to the development project (Heirman and Carysse, 2007). However, Iansiti (1995) argued that process flexibility and responsiveness are key success factors in the chaotic early-stage environment. Correspondingly, recent research has begun to explore the potential negative impact of gated processes on innovation (Sethi and Iqbal, 2008). For innovative new ventures, there is a need for some structure without the burden of arduous stage-gates or control systems. Stated formally:

Proposition 2. New ventures will avoid formal processes and large stage-gate methodologies.

Predevelopment market planning. Market research planning for effective NPD is well-established for mature corporations (Cooper, 2001; Freeman, 1982; Rothwell, 1985). Still, Cooper (2005) recently found that only a quarter of the companies in his sample rigorously executed market planning at the front end of projects. He found that three quarters of the products launched by companies who did rigorous market planning were successful. This market planning includes careful market segmentation, identifying different types of users and buyers, and understanding the perceived and latent needs of both (Urban and Hauser, 1993). The importance of marketing in new ventures was shown in Song et al.'s (2008) meta-analysis, which found that experience in marketing was a significant success factor. For early-stage firms' developing physical products, emphasis on market planning at the front end of new product development may be positively associated with success factors

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such as project breakeven timing and cumulative sales overtime—in line with established firm research. However, the reality of what can be accomplished given resource constraints can be telling. This research looks at three of Cooper's (2001) critical marketing factors in this investigation that best represent up-front planning and "homework." First, do these firms employ formal marketing methods? This includes the use of formal techniques such as surveys, focus groups, and detailed interviews with potential customers in order to ascertain perceived and latent needs. Second, this paper looks at how much "up-front homework" was performed by the development team. If formal marketing data are collected—do they analyze them? How rigorous is the analysis? For the resource-constrained new firm, the cost and time to perform extensive market research and analysis can be limiting, given limited manpower and budgets. Finally, the paper looks at the usage of formal marketing and launch plans during the development process. Are these firms executing to predeveloped plans? Alternatively, are they "flying by the seat of their pants" in the rush to get something to market? In committing resources to predevelopment marketing and planning, the new venture will need to maximize the efficiency of research within the constraints of budget time and dollars. For example, the small firm may decide to speak informally to a small group of potential users rather than pay for expensive focus groups. Stated formally:

Proposition 3. New ventures will not perform traditional market research and planning.

In the next section, research methodology and data collection techniques are reviewed.

Research Methodology and Data Collection

The objective of this in-depth field study was to explore how and why early-stage companies are developing physical products in relation to established NPD practices. To conduct the research, two firms were engaged, which is generally considered more robust than single case studies (Yin, 1994). The cases were deliberately selected and the case methodology presented here is consistent with the objectives of qualitative research (Glaser and Strauss, 1965; Silverman, 2000). The research methodology follows closely to qualitative works including Karjalainen and Snelders (2010), Brockman et al. (2010), and Mabert et al. (1992) that utilize in-depth small sample size case research to drive new product development theory. Important to this research is to ascertain factors that are influential on practice deployment and outcomes, both internal and external. As such, in addition to company principles, general employees, suppliers, and vendors were interviewed. Our qualitative research included independent factors or variables such as marketing practices and specific outcomes measures such as development cost and timing. The principal investigator (PI) was allowed access to development and managerial meetings, detailed financial data, and outside contacts.

To reduce any potential sources of bias, multiple data sources were used (multiple interviews, review of internal documents, and e-mails, etc.) to minimize interpretive problems. In-depth interviews and data collection were performed over a six-year period. Firm A was investigated from 2003 to 2007, and Firm B was investigated from 2001 to 2004. It shall be noted that although the study spans six years, the case time frame is considered to be common and does not contain bias. The engineering technology and communication methods used to develop the projects were consistent among both firms and time periods (computer-aided design [CAD] methods, communication methods, etc.) (Buyukozkan, Dereli, and Baykasoglu, (2004). The PI joined planning and engineering meetings, taking written notes. Vendors and subcontractors were engaged in the research to capture marginal persons and aspects of events. For in-depth research, a research guide and questions were developed to channel data collection and discussion during the project (these can be found in the Appendix). These were used to interview the founders, employees, and vendors. Insights into team experience, interaction, communication, and project management were gathered along with more specific information, such as software tools used, tooling costs, and prototyping methods. Over 10 in-depth interviews were conducted at each firm. Data collection was standardized to increase validity of comparisons between the two case firms. Interviews varied from 30 minutes to over an hour. Interviews were transcribed and/or recorded. Finally, a senior academic advisor was debriefed regularly by the PI to track research progress.

Sample

Two firms producing physical, mechanical consumer products were selected. As is common in field research, the criteria of the desired sample firm was established, then firms that fit the desired criteria were identified and selected (Silverman, 2000). The selection criteria included industry and product type, intellectual property, and funding source. As Wilcox (2008) notes in product development field research in the medical field, "as with any research, validity is enhanced to the extent that the

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research sample accurately reflects the population of interest." Evidence collected that is relevant to the research questions of the study is reported—and where possible—the independence of the results has been verified. Both firms agreed to participate in the study because of general interest in collaborating with a large research university. No compensation or incentives were provided by the PI. Both firms are disguised to further enhance data reporting validity. These firms were deemed appropriate because each complied with our selection criteria and were actively in the midst of planning development of their products. In terms of industry selection, both of the study firms developed a multipart product that requires assembly. Their development included the design and analysis of components that needed to be sourced and manufactured, thus increasing supply chain complexity over other types of new ventures, such as software or Internet firms. Heirman and Carysse (2007) found that service and software firms' NPD practice implementation can be markedly different than those that develop physical products.

It can be argued that physical product companies best relate to established firm NPD literature because of the variety of team members and development steps required from concept development through product testing, launch, and commercialization. Because the work of Cooper (2001, 2005) is used as a research guide, selecting products where marketability and user interaction are important, products were chosen that are primarily sold via retailers (business-to-consumer). These products must have a good value proposition, and meet or exceed customer expectations in terms of design and functionality. Additionally, these types of products best correlate to existing NPD research regarding predevelopment market planning and user-centered design (Cooper, 2001; Norman, 2002). Another criterion in our study was intellectual property; firms were selected that at least have a patent pending status on both the functional utility and design of their products. A hallmark of successful technology-based new ventures is the development of innovative and patentable technology (Rosenkopf and Nerkar, 2001). Firms that required substantial investment to reach the market from either angel or venture capital investors were controlled. Investment ranges from $100,000 to several million dollars were selected. Firm size was also controlled, as firms with fewer than 20 employees were selected. Firms with fewer than 20 employees represent 90% of small and early-stage firms (U.S. Small Business Administration, 2004). The average size of a high technology firm has been found to be eight people (Friar and Eddleston, 2003). Lastly, firm age was controlled. Following VanderWerf and Brush's (1989) recommendations, the sample firms' size was restricted to firms less than 10 years old. In order to reduce variation of results, data collection focused on the first product only, and not successive variants where development processes or procedures might be modified or enhanced. Company profiles are discussed in the following subsection.

Company profiles. Firm A is a designer and marketer of proprietary office machines (U.S. industry classification 3579). Their products are designed in the United States, manufactured in Asia, and sold in major retail outlets worldwide. The company was founded in 2003 by a 30-year-old first-time entrepreneur who conceived the product idea after being injured using a conventional office product. After some additional research, he decided to develop the idea into a commercialized product. Two partners, working virtually on the project, developed the product in less than a year. Total team size at peak development was four, including part-time personnel and vendors. Firm A's first product was developed in 2003 and was launched in early 2004. The firm focused heavily on intellectual property protection of both the function (utility patent) and form (design patent). Several hundred thousand dollars of angel investment was raised, including in-kind engineering from one of the core team members. Sales revenue grew to over $2.5 million per year in the third year of operation—fostered by successive variants based on the initial product concept. By the end of the study period, the firm had 15 full-time employees.

Firm B designs and markets consumer hand tools (U.S. industry classification 3423). Their products are designed and manufactured and sold in retail outlets primarily in the United States. Two co-founders started the company in 2001. The product idea was an afterthought to both cofounders, only entering development after a more complex technology product failed to garner significant investment. Product design and development took place in less than a year. Team size during peak development was composed of seven. Both utility and design patents were sought on the product. Angel investment of several hundred thousand dollars was used to develop the product, tooling, and inventory. At the end of the third year of operation, the company had four full-time employees. Third-year sales revenue based on product sales was approximately $1 million. Both of these firms can be considered *gazelles*—fast growing small firms that create approximately 70% of net new jobs in the United States. (U.S. Small Business Administration, 2004; Zimmerer and Scarborough, 2008). A summary of company profiles can be found in Table 1.

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Table 1. Company Profile Summary

	Firm A	Firm B
Industry	Business-to-consumer product—U.S. Industry Class 3579	Business-to-consumer product—U.S. Industry Class 3423
Product	Multipart, assembled product	Multipart, assembled product
Primary sales channel	Retail	Retail
User interaction	Direct use of product by consumer	Direct use of product by consumer
Intellectual property	At least patent pending, both design and utility	At least patent pending, both design and utility
Funding source	Angel investment for equity	Angel investment for equity
Development team size (1st product)	4	7
Company founding date	2003	2000
Number of founders	2	2
Annual revenue after year three	$2.5 M	$1 M

Results

The case study results are discussed in the next section.

The Use of a Cross-Functional Organization

Cooper (2001, 2005) and Griffin and Hauser (1996) have found that in large corporations, a cross-functional organization can have an impact on project cycle time and ultimate product performance. In practice, these organizations have team members from different functional areas (e.g., engineering, sales, marketing, manufacturing) working on a project under a project manager. The intent is that these project teams foster a sense of co-mingling among team members, removing feudal barriers that inhibit innovative solutions (Cooper, 2001). For the two firms in this study, the project teams were small—comprising the founders and several technical engineers designing the product. No dedicated sales, marketing, or financial personnel were on the team—the core team instead performed these functions themselves.

Once the founder of firm A developed his initial idea, he needed to find the appropriate resources to develop the technology and product (his background was in finance and sales). Through networking, the founder was able to make a contract with an inventor whose prior experience involved development of similar tools. The inventor's technology was licensed to a major manufacturer in the early 1990s. The inventor put the chief executive officer (CEO) in touch with an engineer that became the first employee and project manager. He had direct experience with the technology from a previous position at the firm who licensed the original patents. Firm A's project development team was based in Philadelphia and Los Angeles. The development core team consisted of (1) an inventor with 22 years of development experience including 10 years with similar tools, and (2) a lead engineer with 5 years of development and manufacturing experience, including 3 years with similar office tools. The other core team member was the company founder/CEO, who primarily handled fundraising and sales leads. Contracted part-time manufacturing representatives based mainly in Taiwan joined the team later during development. The initial development team consisting of the outside inventor and lead engineer communicated daily using e-mail and phone. During initial development, the team did not meet in person—as all communication was done remotely. The team did meet in person during the final stages of development when tooling debugging began.

Firm B's founders had met in 2000 and worked extensively on a high-technology communication device that ultimately did not obtain venture capital funding. Consequently, the founders agreed to take any remaining start-up funds and develop an unrelated product. The only internal core team member during development of the new product was one of the founders; as such the entire development team was virtual and part time. This included engineering, manufacturing, industrial design, prototype, and component vendors. The cofounder acted as project manager, instructing the external team members to perform pertinent tasks when and where needed. For example, in weekly teleconferences, the cofounder would review engineering progress on CAD and facilitate input from the designers in Los Angeles. At no time during development did the entire team meet in person. The team's project manager had 5 years of design and development experience, the mechanical engineer had 15 years of experience, the CAD designer had five years of modeling experience, and the industrial designer had 10 years of experience. The main forms of communication were e-mail, phone, and fax. As mentioned, the team communicated weekly via phone conferences to discuss design iterations.

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With regards to a cross-functional organization in large, established firms, there are similarities and differences when comparing to the firms in this study. Cooper (2001) suggests that firms cut across functional barriers and form cross-functional project teams led by a project manager. In this sense, both teams exhibit characteristics of a traditional team. They were both led by a project manager leading the technical team members from concept to production. In Firm A, each of the founders played several concomitant roles in the organization. The CEO's primary functions were fundraising and sales—engineering and technical decisions were left to the engineering team. He was, however, involved in the design process at one critical point, namely the decision whether or not to proceed to production tooling. The lead engineer had multiple "hats" as well, interfacing with design and engineering, and aiding the CEO where needed in sales and fundraising. The Los Angeles-based inventor was on the team part time as a contractor who was paid for his engineering hours through an equity position in the company. A Taiwan-based manufacturing representative came on the team part time in the last several months of the project as tooling development proceeded. Similar to Firm A, Firm B's CEO was primarily focused on sales and fundraising. The project manager, again similar to Firm A, played multiple roles in the organization, leading the design and manufacturing efforts, but also aiding the CEO where needed. The development team, a matrix of outside resources, was given autonomous decision-making power on the design and successive iterations.

Besides having a project manager and resources committed to the project, similarities to Cooper's description of cross-functional team are scant. Large corporations have multiple departments to draw upon, with their own functional expertise. In the start-up environment, it is the founders and a limited number of technical resources that are driving the project. Multiple functions need to be performed, not by members of different departments but by the team itself. This is a clear distinction between large organizations and the new venture. For example, if a sales sheet needed to be developed with pricing, packaging information, etc., team members would be assigned quickly. Team members that had no experience with sales or marketing would "jump" on the task, learn, and "get the job done." Instead of relying on functional departments to complete tasks, these small project teams would assign tasks to wherever a resource opening permitted, regardless of expertise. One team member from Firm B stated: "I love doing things other than engineering on this project—I have never worked on packaging before—it's actually exciting to learn something new." This methodology also allowed the project to keep progressing without dedicated resources or waiting for them—by allocating *what you had* to *what you needed*.

Another observation of both firms was that the project manager, CEO, and development team acted extremely quickly on decisions. Often, critical design issues were solved during a teleconference, with limited issue vetting outside the confines of the meeting. Other tasks, outside of core engineering, were handled in the same fashion. As the project manager of Firm A stated: "we need to get things done to get this product to market, we just go out and do it—even if we have to learn how." Lastly, both firms relied heavily on part-time outsourced team members—another departure from Cooper's (2005) definition of having full-time members. With regard to support of the propositions, Proposition 1 was supported. Neither firm used *traditional* full-time cross-functional teams as defined by literature. However, both firms' approaches to teams and how to organize personnel and tackle problems were extremely similar. Lastly, both firms had a large percentage of team members work for equity, which has a direct impact on cash flow and project motivation. A summary of both firms' team characteristics are shown in Table 2.

The Formality of the NPD Process

Cooper (2001, 2005) and McGrath (1995) have popularized the use of formalized gated development processes, and large established firms worldwide have embraced these types of operational methods. The impetus, as Cooper (2001) states, is that an innovation program without a process is like players on a football field without a game plan or plays. For the small new venture, moving the "ball down the field" is all-important, as every minute that passes without moving closer to production draws down available funds. For established firms, planning and following a defined methodology is deemed as critical for successful commercialization.

For the new ventures in this study, the research framework evaluated three criteria of formalized development processes in which to compare and draw insight versus best practice at large firms. The first is a process itself. Firm A did not use a written formalized process during development. Instead, they relied on the experience of the team to guide development and decisions. It is interesting to note that there was little CEO involvement during the project. The main interaction with the development team was limited to periodic updates from the lead engineer and a final decision among team members to move

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Table 2. Cross-Functional Team Adoption Summary

Best Practice	Critical Characteristics	Firm A	Firm B
Cross-functional team	Use of traditional, full-time teams	No. Two full-time founders. All other resources were part-time contractors.	No. Two full-time founders. All other resources were part-time contractors.
	Use of part-time external resources	Yes. Part-time external resources used for design, engineering, and manufacturing.	Yes. Part-time external resources used for design, engineering, and manufacturing.
	Work performed only when resources in place	No. Work driven by goals—internal team members would fill responsibility gaps where needed.	No. Work driven by goals—internal team members would fill responsibility gaps where needed.
	Multiple roles played by founders	Yes. Founders switched between marketing, sales, finance, engineering, and project management.	Yes. Founders switched between marketing, sales, finance, engineering, and project management.

forward with tooling development and launch after testing was completed. This occurred six months into the project. Other than e-mail tasks lists and Microsoft Excel spreadsheets to track costs-of-goods-sold (COGS), tooling and launch issues, no formal project management tools were used to guide development.

As with Firm A, Firm B used no formal development process. Objectives were discussed in weekly teleconferences, and progress to tasks was tracked using e-mail and project spreadsheets (Microsoft Outlook and Excel). These tools were used from the beginning of the project until launch. Informal management decisions between the founders regarding continuing development were performed at two points during development (before detailed engineering and before tooling kick-off to ensure funds were in place). For these decisions, no defined metrics were used to evaluate the project's performance—other than if the technology worked and if it could be produced economically.

In comparing the innovation process of these firms versus large, established firm practices, there are substantial differences between the two. Although both firms did have informal decision points, both did not use a formal process. No written procedures were used nor were development meetings run to specific deliverables or defined steps. The teams relied on experience of the project team, specifications of the product, and pressure to commercialize as main drivers during the project. In observing design meetings, the objectives were clear of what needed to be done before the next meeting, and all participants would agree to tasks. The project manager in both cases drove the process, rather than working the team to a defined process. Firm B's project manager states:

> *I knew when certain items needed to be completed, and we would work to that. The main goal was to get the product designed and meet the functional specification. Concepts, to prototypes, to testing was done iteratively and quickly to get to production. All other items (packaging, sales, etc.) that could be done in parallel were left to be completed during production ramp—this allowed maximum focus on the project tasks at hand. 50% of my time at my last job (automotive manufacturer) was spent filling out project paperwork for the development process and meetings—that has no place here.*

Cooper (2001) notes that formal project control is critical; in these cases, all controls were informal. Meetings were run informally; and project management items such as Gantt charts were not used.

Proposition 2 was supported. Both firms did not use written procedures but both used informal project management tools to some extent—Firm A being more limited in their use than Firm B. Both firms did not use a formal gated process, but both did have an informal decision point at some point during development. However, both firms did have a team member function as a project manager, although use of traditional project management tools was nonexistent or limited. A summary of both firms use of a formal process is shown in Table 3.

The Use of Up-Front Market Planning

A hallmark of Cooper's research with large, established firms notes the importance of up-front "homework" and market research. Both of the firms in this study designed and produced consumer products in which pricing, design appeal, and functionality are key drivers for success (Norman, 2002). It would seem logical that similar emphasis on early market exploration and translation into defined product parameters would lead to analogous outcomes.

For Firm A, some predevelopment benchmarking was performed. This included evaluating the most common

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Table 3. Formal Process Adoption Summary

Best Practice	Critical Characteristics	Firm A	Firm B
Use of a formal process	Use of a formal, written process	No. No formal, written, and prescribed process used. Process driven by project milestones.	No. No formal, written, and prescribed process used. Process driven by project milestones.
	Formal gated process	No. Informal team collective decision to more forward with commercialization.	No. Two informal meetings held between the founders to decide whether to move forward with detailed design and tooling.
	Use of project control (project management)	Limited. Founder played role of project manager. Other than e-mail follow-up on tasks, no formal project management tools were used.	Limited. Founder played role of project manager. Primary tools used were e-mail and spreadsheets to track project goals.

office tools in the United States and Europe for comparative function, design, and cost. Then, the unit was designed to "fit in" with traditional office tool appearance parameters so that it would be recognized functionally, and not alienate potential consumers with an unfamiliar design. The selling price was determined by placing the office tool at the top of the selling range because of its uniqueness and functionality. During the later stages of design, prototypes were fabricated and tested by the core team in-house with competitive tools to verify critical functional specifications. Opinion of the development team drove design tweaks and specifications. Potential customers or resellers performed no external product evaluation or testing. Additionally, no formal tracking of customer needs relating to engineering specifications was completed. Formal discussions with retailer outlets and potential customers were not performed. The company also did not develop a formal launch and commercialization plan. Once manufacturing capability and capacity—and resulting costs/margins—were finalized, orders for the product were taken and the product was shipped. All focus of the firm shifted from development to maximizing sales to generate profit.

Firm B had a similar approach. During the initial stages of development, market analysis on competitive firms and product benchmarking research were performed concurrently with product design. Leading manufacturers were identified, and competing products and associated price points were evaluated. The primary form of research included Internet searches, in-store visits, and competitor evaluation. Product attributes were developed by the founders in response to research and general frustration with current product offerings. During the later stages of the design effort, a 600-respondent survey was undertaken with potential customers. This information was used to shape business plan opportunity assessment and legitimize product attributes and potential retail pricing for prospective investors and resellers. However, the survey results were analyzed post hoc, and did not guide product design. Some formal planning and launch analysis did take place in the guise of business plans to attract potential investors.

In comparing these firms' predevelopment marketing to larger firm practices, there are clear differences in how the work was performed. Both firms used very informal benchmarking and market investigations. In both cases, potential consumers were not involved in the design process. Defined focus groups, house of quality, conjoint analyses, etc. were eschewed for a very basic definition of functionality and design needs, driven by the opinion and experience of the core team who acted as lead users. Firm A's CEO stated, "We knew if we developed a product that could work better with a new technology—and have a competitive price point—people would buy it." Firm B's cofounder said, "My partner and I sketched out the design and functionality requirements in a few hours, based on our experiences and frustrations—we kind of designed the product for ourselves." In both cases, the firms did not have a dedicated commercialization and launch plan—at most each had an informal section in the company business plan noting sales strategy.

Proposition 3 was supported. Although both firms performed limited benchmarking and competitive analyses, they did not actively utilize retailer, sales, or customer input during development. Up-front homework took a back seat to experience and instincts of the core team. Lastly, both firms did not use formal commercialization and launch plans. A summary of the investigation is shown in Table 4.

Epilogue and Outcomes

For their first product, each of the firm's projects ended up being similar in level of total development cost and

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Table 4. Up-Front Market Planning Adoption Summary

Best Practice	Critical Characteristics	Firm A	Firm B
Use of a formal process	Use of formal marketing methods	No. The core team investigated the competitive space and designed the product to internal specifications—which exceeded those of current offerings.	No. The market was investigated informally. Competitive products were evaluated and feature sets were developed from an internal lead-user focus.
	Use of up-front analysis and planning	Limited. Simple market and product/price investigations were used. The core team acted as lead users.	Limited. Simple, low-cost investigation used. Surveys were used after design to validate business opportunity for investment.
	Use of a formal launch/commercialization plan	No. Firm focus was on development and manufacturing. Once the product was ready, all effort shifted to sales.	Limited. Some planning was used, primarily in business plans for prospective investors.

time to market (less than $100,000 and under a year). Both firms aggressively pursued extremely low-cost methods and "design for equity" arrangements. For example, Firm A had the design work completed for equity. The manufacturer also invested in the project by amortizing all tooling costs, reducing up-front cash expenditure greatly. Firm B also relied heavily on "sweat equity," in-kind investment, and state grants. Both firms estimated that under normal development conditions, total costs would have been at least multiple times higher. In addition, each of the projects resulted in successful completion and market introduction, which was unexpected at the beginning of the study because of the inherent risk in NPD in start-up firms. At the time of writing, both companies have continued to grow and expand their product offerings. In both cases, the first product developed is still in production and is experiencing consistent year-over-year sales growth. Both firms' products have received national attention for their design and functionality. A summary of company outcomes is shown in Table 5.

Table 5. Epilogue—Firm Outcomes

	Firm A	Firm B
Time-to-market	10 months	9 months
Development cost	$10,000	$50,000
Internal development person hours	2,120	900
Tooling cost	$85,000	$37,500
Product gross margin	50%	39%
Project breakeven timing	<1 year	<1 year
Cumulative sales (first three years)	>100,000 units	>100,000 units
As of 2010, product still on sale	Yes	Yes
Product received national press	Yes	Yes

Implications, Limitations, and Future Research

This study contributes to academic research and practitioners on several fronts. The first is that evidence is shown that start-ups are different in their approach to and adoption of NPD best practices, which has several theoretical implications. Using Cooper's (2001) work as a proxy, the use of commonly prescribed practices such as cross-functional teams, formal NPD processes, and up-front market planning was investigated in detail. Largely, the differences in development approach from what is prescribed for large, established firms are driven by necessity. These new firms simply did not have the resources (financial or human) to create multi- or cross-functional teams or organizations in the traditional sense for their first product. Instead, both firms used virtual teams that brought functional expertise into the project as needed rather than have them involved from the beginning. This is much different from the large organization where dedicated, full-time personnel are attached to projects. As is common among start-ups, the firms in this study had two founders—amassing a large full-time team would have drained scant finances quickly. These firms sought a critical path, driven by financials and the need to commercialize. In our research, most of the team members worked for equity, further reducing capital cost structure. The team members themselves played multiple roles during development—from design to sales—not relying on functional departments or outside personnel (other than engineers). Notably, both firms kept development decisions and design input inside the core team, relying on their own instincts and skill. Time was critical, and they had to economize their time and focus on key

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development milestones rather than reporting to phase metrics. As noted in Brockman et al. (2010), NPD teams may obtain motivation at specific milestones, creating a strong atmosphere of commitment. The difference between large firms and the start-up space is that the motivation is keeping the firm alive, not sinking into premature bankruptcy, and the potential monetization of earned equity. This contributes to an intrinsic motivation within the core team.

The NPD process used by both firms was informal—much more skeletal than the structured processes that are recommended by researchers such as Cooper (2001). The study found that these firms put less focus on managing the process and more emphasis on managing their goals (the main driver being getting the first product to market). In comparing the innovation processes of these firms compared to large, established firms, several substantial differences are noted. In addition to little or no written procedures being used, the investigation found that development meetings did not run to specific paper-based deliverables or defined steps. The main drivers of projects were the teams' experiences, specifications of the product, and pressure to commercialize. In observing design meetings, the goals were clearly stated. Teams understood what needed to be done before the next meeting, and all participants agreed to tasks. The project manager or "heavyweight" drove the process rather than working to a process to be reviewed by management. In terms of paperwork and tools, other than e-mail task lists and Microsoft Excel spreadsheets to track costs-of-goods-sold (COGS), tooling, and launch issues, there was little or no formal project management tools used to guide development. Lastly, the main interaction of upper-level management (e.g., board members or investors) with the development team was limited to periodic updates from the lead engineer and a final decision gate (go/no-go) meeting that approved moving forward to tooling development and launch.

In comparing our study firms' predevelopment marketing to larger firm practices, there are clear differences in how the work was performed. The new firms in our research used very informal benchmarking and market investigations. Potential consumers were not involved in the design process. Marketing and up-front "homework" was done primarily inside the core team. What drove a project to completion was relying on team experience or a "learn as you go approach." Again, the driver for this type of truncated market research approach was the lack of resources and need to increase the project's speed-to-market. The ideas for product specifications were usually generated from experience within the field; team members identified "gaps" in the current marketplace. In a sense, core team members were themselves savvy, experienced end users, and they were willing to market new ideas based on their own empathetic point of view. Very often, we witnessed teams designing product to their own high standards and not to those of their loyal end users. The opinion and experience of the core team was critical to define project parameters and specifications. Even then, teams primarily focused on a very basic definition of functionality and design needs. In many ways, these two firms approached conceptual development and market research very similarly to industrial design firms (e.g., IDEO and Continuum), where highly experienced, small teams develop products by immersing themselves into the problem space, iterating concepts and prototypes, and quickly realizing a final solution (Meyer and Marion, 2010).

What makes these findings of differences in NPD adoption even more interesting is the fact that both of these firms exist in a space of "traditional NPD." Both firms developed and introduced an incremental innovation, where there are existing markets, customers, and competition. If any new venture that were to be similar in their approach to Cooper's methods, it can be argued it would be these firms. Yet the study found substantial differences in their approaches. On first observance, these firms fit Leonard-Barton and Boyle's (1994) typology of current markets, where the need is known, and there is room for an improved solution. Leonard-Barton and Doyle found these types of firms tended to use traditional marketing and product development techniques. Traditional NPD approaches do not fully address the unique attributes of the new firm environment of limited resources and the need to successfully commercialize or fail. It can be argued that for new ventures—even those operating in existing market spaces—it is inadequate to simply truncate existing methodologies (e.g., Stage-Gate Xpress or Lite, etc.) and expect them to be well suited for efficient and effective NPD. The upshot of these findings is that NPD processes in start-ups should be studied as a different and important context separate from large company NPD.

The second major implication of this research is the potential lessons that can be drawn from product development methods at small, new ventures for large companies. The first lesson is empowerment. Both firms in this study had founders, CEOs, and investors that gave the development team near total decision-making power over product functionality and design. This not only energized and empowered the team but also made development decisions extremely efficient. Second, although the teams

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were small, the scope of the projects was substantial (e.g., the need for sales, manufacturing, packaging, supply chain management, etc.). This necessitated that team members fill roles that are outside of their expertise, and they were often excited about the prospects and challenges of working on something new. Third, both of these firms have been highly successful, and used very little of Cooper's (2001) critical success factors in any substantive, rigorous manner. At best, the core team would discuss items and quickly make decisions about functionality, price points, etc. Little time or resources were spent on the process; all focus was on moving the project forward. The result was that these firms adopted a flexible development strategy working through an informal process. Certain practices (such as benchmarking competitive products) were used when they were needed to a level that was appropriate given the needs of the product and management team. This supports Iansiti's (1995) argument that process flexibility and responsiveness are key success factors for NPD in large companies in the chaotic early-stage environment. This also indicates further support for the development of more flexible, simplified development procedures (cf., Sethi and Iqbal, 2008).

Given the limitations of the study presented here, additional research is needed to soundly quantify the impact of NPD practices on start-up firm success and to solidify differences between small new ventures and large corporations—and develop unique methods for the upper right quadrant of new markets and new ventures. Cooper's (2001, 2005) work focuses on limiting product failure and increasing success. There is a need for additional research in new ventures to: (1) explore the tension between the quest to limit failure versus stifling innovation through onerous process; (2) develop managerial insights and practices for both spaces of firms; and (3) understand the factors that influence the adoption of more rigorous frameworks as start-ups grow and evolve. This study is a first step toward that end. While limited in the ability to generalize results, this research has given clear indications of the distinctiveness of NPD in new ventures, and the potential impact of entrepreneurial firm research and practice for successful commercial innovation. In future work, we plan to continue in-depth case analysis while refining the research methods for large survey-based studies. What is clear is that small, new ventures are different from large corporations, and their NPD methods are different. The two firms in this study continue to grow profitably, but they maintain informal processes with sparse adoption of commonly prescribed best practice. Entrepreneurship and NPD researchers would be well served to make this area a central tenet of future study—as understanding and replicating successful innovation is key to our economic growth.

References

Acs, Z. J., P. Arenius, M. Hay, and M. Minniti. 2004. *Global Entrepreneurship Monitor 2004 executive report*. Babson Park, MA: Babson College, Ewing Marion Kauffman Foundation, and London Business School.

Adams, M. 2004. PDMA Foundation New Product Development Report of Initial Findings: Summary of Responses from 2004 CPAS. *Product Development and Management Association*. Available at http://www.pdma.org/shop_pdma_description.cfm?pk_store_product=25.

Adams, R., H. Almeida, and D. Ferreira. 2009. Understanding the relationship between founders-CEOs and firm performance. *Journal of Empirical Finance* 16: 136–50.

Adler, P. 1995. Interdepartmental interdependence and coordination: The case of the design/manufacturing interface. *Organization Science* 6 (2): 147–67.

Balachandra, R., and J. Friar. 1997. Factors for success in R&D projects and new product innovation: A contextual framework. *IEEE Transactions on Engineering Management* 44 (3): 276–87.

Barczak, G., and D. Wilemon. 2003. Team member experiences in new product development: Views from the trenches. *R&D Management* 33 (5): 463–79.

Brockman, B. K., M. E. Rawlston, M. A. Jones, and D. Halstead. 2010. An exploratory model of interpersonal cohesiveness in new product development teams. *Journal of Product Innovation Management* 27: 201–19.

Buyukozkan, G., T. Dereli, and A. Baykasoglu. 2004. A survey on methods and tools of concurrent new product development and agile manufacturing. *Journal of Intelligent Manufacturing* 14: 731–51.

Carree, M. A., and A. R. Thurik. 2003. The impact of entrepreneurship on economic growth. In *Handbook of entrepreneurship research*, ed. D. B. Audretsch and Z. J. Acs, 437–71. Boston/Dordrecht: Kluwer Academic Publishers.

Cooper, A. C. 1970. The Palo Alto experience. *Industrial Research* May: 58–60.

Cooper, R. G. 2001. *Winning at new products*. Cambridge, MA: Perseus Publishing.

Cooper, R. G. 2005. *Leadership: Pathways to profitable innovation* (2nd ed.). New York: Basic Books.

Eppinger, S. D., and A. R. Chitkara. 2006. The new practice of global product development. *MIT Sloan Management Review* 47 (4): 22–30.

Freeman, C. 1982. *The economics of industrial innovation*. Cambridge, MA: The MIT Press.

Friar, J. H., and K. Eddleston. 2003. What it takes to succeed depends on small business type: A contingency typology for small businesses, *Proceedings of the Academy of Management*, Seattle, WA, August 4, 2003.

Friar, J. H., T. J. Marion, and R. Kinnunen. 2009. Adnexus corporation case. *Proceedings for the NACRA* Conference, *October 29–31, 2009*, Santa Cruz, CA.

Gartner, W. B. 1985. A conceptual framework for describing the phenomenon of new venture creation. *Academy of Management Review* 10 (4): 696–706.

Gartner, W. B., J. A. Starr, and S. Bhat. 1998. Predicting new venture survival: An analysis of "anatomy of a start-up" cases from Inc. Magazine. *Journal of Business Venturing* 14 (2): 215–32.

Glaser, B. G., and A. L. Strauss. 1965. The discovery of substantive theory: A basic strategy underlying qualitative research. *The American Behavioral Scientist* 8 (6): 5–12.

15405885, 2012, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1540-5885.2012.00930.x by Bahria University Karachi, Wiley Online Library on [27/09/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Griffin, A., and J. R. Hauser. 1996. Integrating R&D and marketing: A review and analysis of literature. *Journal of Product Innovation Management* 13 (3): 191–215.

Headd, B. 2003. Redefining business success: Distinguishing between closure and failure. *Small Business Economics* 21 (1): 51–61.

Heirman, A., and B. Carysse. 2007. Which tangible and intangible assets matter for innovation speed in start-ups? *Journal of Product Innovation Management* 24: 303–15.

Iansiti, M. 1995. Shooting the rapids: Managing product development in a turbulent environment. *California Management Review* 38 (1): 37–58.

Jackson, P. 2000. Ethnography. In *Cultural geography: A critical dictionary of key concepts*, ed. R. Johnston, D. Gregory, G. Pratt, and M. Watts, 103–9. Oxford: Blackwell.

Karjalainen, T., and D. Snelders. 2010. Designing visual recognition for the brand. *The Journal of Product Innovation Management* 27: 6–22.

Leonard-Barton, D., and J. L. Boyle. 1994. Commercialing technology: Imaginative understanding of user needs. In *Engines of innovation: U.S. industrial research at the end of an era*, ed. R. S. Rosenbloom and W. J. Spencer, 182–221. Boston, MA: Harvard Business School Press.

Loch, C. H., M. E. Solt, and E. M. Bailey. 2008. Diagnosing unforeseeable uncertainty in a new venture. *Journal of Product Innovation Management* 25 (1): 28–46.

Mabert, V. A., J. F. Muth, and R. W. Schmenner. 1992. Collapsing new product development times: Six case studies. *Journal of Product Innovation Management* 9: 200–12.

Marion, T. J., and M. A. Meyer. 2011. Applying industrial design and cost engineering to new product development in early-stage firms. *The Journal of Product Innovation Management* 28: 773–86.

McGrath, M. E. 1995. *Product strategy for high-technology companies*. New York: Irwin Professional Publishing.

Meyer, M. H. 2007. *The fast path to corporate growth: Leveraging knowledge and technologies to new market applications*. New York: Oxford University Press.

Meyer, M. H., and A. P. Lehnerd. 1997. *The power of product platforms*. New York: The Free Press.

Meyer, M. H., and T. J. Marion. 2010. Innovating for effectiveness: Lessons from design firms. *Research-Technology Management* 53 (5): 21–28.

Mol, M. J. 2001. Ford Mondeo: A Model T World Car? Published dissertation, Eramus University Rotterdam, The Netherlands.

Nelson, R. R. 1991. Why do firms differ, and how does it matter? *Strategic Management Journal* 12: 61–74.

Norman, D. A. 2002. *The design of everyday things* (2nd ed.). New York, NY: Currency/Doubleday.

Oakly, R. P. 2003. Technical entrepreneurship in high technology small firms: Some observations on the implications for management. *Technovation* 23(3): 679–88.

Price, R. W. 2006. *Mastering entrepreneurship, entrepreneurship* (5th ed.). Dubuque, IQ: McGraw-Hil.

Project Management Institute (PMI) Standards Committee. 1996. *Guide to the project management body of knowledge*. Upper Darby, PA: Project Management Institute.

Rosenkopf, L., and A. Nerkar. 2001. Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal* 22 (4): 287–306.

Rothwell, R. 1985. Project Sappho: Comparative study of success and failure in industrial innovation. *Information Age* 7 (4): 215–20.

Saxenian, A. 1985. Silicon Valley and Route 128: Regional prototypes or historic exceptions? In *High technology, space and society*, ed. M. Castells, 81–105. Beverly Hills, CA: Sage Publications.

Schoonhoven, C. B., K. M. Eisenhardt, and K. Lyman. 1990. Speeding new products to market: Waiting time to first product introduction in new firms. *Administrative Science Quarterly* 35: 177–207.

Sethi, R., and Z. Iqbal. 2008. Stage-gate controls, learning failure, and adverse effect on novel new products. *Journal of Marketing* 71 (1): 118–34.

Silverman, D. 2000. *Doing qualitative research*. London: Sage Publications.

Smith, P., and D. Reinertsen. 1997. *Developing products in half the time: New rules, new tools* (2nd ed.). Hoboken, NJ: John Wiley & Sons.

Song, M., K. Podoynitsyna, H. van der Bij, and J. Halman. 2008. Success factors in new ventures: A meta-analysis. *Journal of Product Innovation Management* 25 (1): 7–20.

Taguchi, H., and I. Nonaka. 1986. The new product development game. *Harvard Business Review* 64 (1): 146–47.

Ulrich, K. T., and S. D. Eppinger. 2004. *Product design and development* (3rd ed.). New York: Irwin McGraw-Hill.

Urban, G. L., and J. R. Hauser. 1993. *Design and marketing of new products* (2nd ed.). Englewood Cliffs, NJ: Prentice-Hall.

U.S. Small Business Administration. 2004. *Small business by the numbers*. Washington, DC: Author.

VanderWerf, P. A., and C. Brush. 1989. Achieving empirical progress in an undefined field. *Entrepreneurship Theory and Practice* 14 (2): 45–58.

Wheelwright, S., and K. B. Clark. 1992. *Revolutionizing product development*. New York: The Free Press.

Wilcox, S. 2008. Ethnographic research and the problem of validity. *Medical Device & Diagnostic Industry* February, 2008, 56–61.

Yin, R. K. 1994. *Case study research: Design and methods*. Thousand Oaks, CA: Sage Publications.

Zimmerer, T. W., and N. M. Scarborough. 2008. *Essentials of entrepreneurship and small business management* (5th ed.). Upper Saddle River, NJ: Pearson Prentice Hall.

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Appendix A. Case Study Specifics and Guidelines

Study Phase	Case Research Methods and Specifics
Observation	
Method	Note and audio tape
Firm access	One-an-one interviews Access to e-mails Founder interviews Employee interviews Vendor interviews
Depth interviews	
Method	Note and audio tape
Total depth interviews:	26
Interview research guide (outcomes)	
How long did it take to develop the first product?	
How much experience did the development team have?	
How much did the development cost?	
How much did tooling cost?	
How much time and money was spent tweaking tooling?	
How much time and money was spent on prototypes and testing?	
What software was used to design the product?	
What types of prototypes were constructed?	
Was cost modeled during development?	
How was cost modeled?	
How did outsourcing affect the design?	
Did low labor costs affect design decisions?	
Did you use design for assembly or manufacture techniques?	
How did you determine selling price?	
Research guide (specific—example questions for predevelopment marketing)	
Are you familiar with the work of Robert Cooper?	
What differentiates your first product? Be specific.	
Why does your product offer superior value?	
Has the competition responded to your product?	
How did you develop your product's specifications?	
Did you talk to potential customers?	
Did you talk to resellers?	
Did you purchase and evaluate competitor's products?	
Did you use formal methods for capturing customer needs?	
Did you write them down and review them as a team?	
Did you review concept specifications with users?	
Did you change specifications and revise concepts?	
Where these iterations reviewed by potential customers?	
Did you use formal focus groups?	
Or, did you talk to a few users? Or, none at all?	
Did the team look at the total market size and space?	
Did you segment the market?	
If so, was this market analysis planned?	
Did you identify supply chain partners?	
Was a formal launch plan developed?	
Was an informal plan developed?	
How was it implemented?	
Who performed the market investigation?	
Was the role formal or informal? Full or part time?	

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